Md Salauddin, PhD

Research Scientist, Nuclear, Plasma, and Radiological Engineering
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Research Impact: 4,927 Google Scholar Citations | 44 Published Journal Articles | 3 Patents

ACADEMIC PROFILE

Interdisciplinary engineering researcher with 10+ years of experience in self-powered sensing, triboelectric and hybrid energy harvesting, flexible/wearable electronics, smart materials, embedded sensing, and cyber-physical monitoring. My research connects device physics and materials with system integration, real-time data acquisition, and application-driven validation. I have authored or coauthored 44 journal publications, presented at major international conferences, contributed to externally funded research programs, mentored undergraduate and graduate researchers, and served as a peer reviewer and guest editor. My long-term goal is to build an independent academic research program at the intersection of smart sensing, energy harvesting, intelligent instrumentation, and resilient engineering systems.

RESEARCH INTERESTS

Self-Powered Sensors and Triboelectric Nanogenerators • Hybrid Electromagnetic-Triboelectric Systems • Flexible and Wearable Electronics • Smart Textiles • MXene/Polymer Nanocomposites • Embedded Systems and IoT • Cyber-Physical Monitoring • Digital Twin-Enabled Sensing • Biomedical and Industrial Sensing

EDUCATION

Ph.D. in Electronic Engineering **2019**
Kwangwoon University, Seoul, South Korea
- Dissertation: Development of Human Body Induced Vibration Driven Hybrid Energy Harvesters using Halbach Magnet Array
- Supervisor: Prof. Jae Yeong Park

M.Sc. in Electrical and Electronic Engineering **2013**
Islamic University, Bangladesh

B.Sc. in Electrical and Electronic Engineering **2011**
Islamic University, Bangladesh

RESEARCH EXPERIENCE

Postdoctoral Research Associate **Aug 2026–Present**
Department of Nuclear, Plasma, and Radiological Engineering, University of Illinois Urbana-Champaign
- Developing TENG-based self-powered sensing approaches for cyber-physical monitoring of a thermal-hydraulic two-tank research testbed.
- Integrating commercial level, flow, and pressure sensing with high-speed triboelectric signal acquisition for condition monitoring and digital-twin/virtual-replica research.
- Designing steady-state, transient, and fault experiments to compare self-powered sensing signatures against conventional instrumentation.
- Working across sensor hardware, signal conditioning, embedded acquisition, and data-driven analysis to support intelligent monitoring of engineering systems.

Postdoctoral Research Associate
Departments of Mechanical Science & Engineering

and Food Science & Human Nutrition, University of Illinois Urbana-Champaign **Dec 2022–Nov 2024**

- Designed wearable sensing systems integrating flexible sensors, embedded electronics, wireless data acquisition, and signal processing.
- Developed triboelectric sensing platforms for biomechanical monitoring of frailty, fall risk, human motion, and healthcare applications.
- Led sensor fabrication, hardware integration, calibration, reliability testing, and application-level validation.
- Mentored undergraduate researchers on wearable sensing systems using flexible PCBs and Bluetooth-enabled modules.

Postdoctoral Brain Pool Fellow **Sep 2019–Dec 2022**

Advanced Sensor and Energy Research Laboratory, Kwangwoon University, South Korea

- Designed and validated wearable triboelectric and hybrid transducers for human-motion sensing, self-powered electronics, and environmental energy harvesting.
- Integrated MXene, MOFs, electrospun nanofibers, and stretchable polymers to improve electrical output, durability, and sensing reliability.
- Led iterative prototyping, experimental characterization, system integration, publication development, and graduate-student mentoring.
- Collaborated with industry on wearable healthcare sensing applications.

Graduate Research Assistant **Sep 2014–Oct 2019**

Advanced Sensor and Energy Research Laboratory, Kwangwoon University, South Korea

- Developed electromagnetic-triboelectric-piezoelectric hybrid nanogenerators and Halbach-array energy harvesters for low-frequency human motion.
- Performed device design, theoretical modeling, fabrication, electromechanical characterization, and practical demonstrations for wearable and portable electronics.

TEACHING AND MENTORING EXPERIENCE

Research Mentor, University of Illinois Urbana-Champaign Dec 2022-Nov 2024

- Mentored REU and undergraduate students on smart textile fabrication, triboelectric systems, and signal processing.
- Delivered guest lectures on TENGs, energy harvesting, and embedded interfaces.

Senior Design Mentor, Department of Electronic Engineering, Kwangwoon University, South Korea Jan 2015-Nov 2022

- Assisted in labs and lectures for Introduction of Micro/Nano Engineering, Microsystem Packaging Engineering, Electromagnetics, and MEMS Fabrication.
- Led student workshops on COMSOL, FEM analysis, MATLAB, and sensor prototyping.
- Planned and conducted weekly small lecture-style classes every Spring semester for senior design students.
- Mentored capstone projects that resulted in multiple peer-reviewed journal articles and conference proceedings.

RESEARCH GRANTS & FUNDING EXPERIENCE

- Brain Pool Program, National Research Foundation of Korea / Ministry of Science and ICT, Grant 2020H1D3A1A02081280, ~201 million KRW, 4 years. Role: Postdoctoral Fellow.
- Project focus: smart textile/fabric-based hybridized energy harvesters for self-powered sensors and wearable/portable electronics.
- Basic Science Research Program, NRF Korea, Grant NRF-2020R1A2C2012820, ~810 million KRW, 4 years. Role: Research Associate.
- Contributed to additional Korean government-supported projects involving RF MEMS, IoT, and wearable multisensor systems.

TECHNICAL EXPERTISE

- Sensor and harvester design: triboelectric, piezoelectric, electromagnetic, and hybrid transducers; electromechanical testing and optimization.
- Fabrication: electrospinning, solution casting, laser patterning, textile coating, flexible/nanocomposite integration.
- Embedded systems and IoT: wireless sensing modules, real-time acquisition, system integration, event-driven processing.
- Materials characterization: SEM, EDS, XPS, XRD, Raman, FTIR, dielectric analysis.
- Modeling and software: COMSOL Multiphysics, FEMM, MATLAB, LabVIEW, Origin.
- Wearable sensing: gait, frailty, motion tracking, biosignal acquisition, remote monitoring, human-subject deployment.

SELECTED SCHOLARLY HIGHLIGHTS

- 44 journal publications, including work in Advanced Science, Advanced Energy Materials, Advanced Functional Materials, Nano Energy, Applied Energy, ACS Applied Materials & Interfaces, and related journals.
- 14 conference presentations, including IEEE MEMS, Transducers, PowerMEMS, IEEE NEMS, NanoKorea, and the Design of Medical Devices Conference.
- 3 Korean patents in electromagnetic and hybrid energy harvesting technologies.
- 6 journal cover features associated with work on triboelectric, hybrid nanogenerator, and self-powered sensing systems.

JOURNAL PUBLICATIONS (44)

1) Y.Y. Choi, P. Karua, M. Salauddin, and L. Cai, Intrinsic Coupling of Radiative Cooling and Triboelectric Responses in Dual-Function ZrO2 Nanocomposites for Adaptive Thermoregulation, **Advanced Science**, (2026), pp. e76121

2) S. Sakib, H. Kun Woo, P. Karua, M. Salauddin, W. Zheng, L. Cai, and J. He, Sustainable production of nitrogen-doped hierarchical porous carbons from fruit Peel wastes for high-performance CO2 capture, **Separation and Purification Technology**, Vol. 380 (2025), pp.135501

3) Y. Young, K. Zhou, H. K. Woo, D. Patel, M. Salauddin, and L. Cai, Radiative Cooling Smart Textiles with Integrated Sensing for Adaptive Thermoregulation, **ACS Materials Letters**, (2024), pp. 4624−4631

4) M. Salauddin, SM S. Rana, M. Sharifuzzaman, H. S. Song, M. S. Reza, S.H. Jeong, and J. Y. Park, Highly Electronegative V2CTx/Silicone Nanocomposite-Based Serpentine Triboelectric Nanogenerator for Wearable Self-Powered Sensors and Sign Language Interpretation, **Advanced Energy Materials**, (2023), pp. 2203812

5) SM S. Rana, M. T. Rahman, M. A. Zahed, S. Lee, Y.D. Shin, S. Seonu, D. Kim, M. Salauddin, T. Bhatta, K. Shresta, and J. Y. Park, Zirconium metal-organic framework and hybridized Co-NPC@MXene nanocomposite coated fabric for stretchable, humidity-resistant triboelectric nanogenerators and selfpowered tactile sensors, **Nano Energy**, Vol. 104 (2022), pp. 107931

6) SM S. Rana, M. Salauddin, M. Sharifuzzaman, S. Lee, Y.D. Shin, H.S. Song, S.H. Jeong, T. Bhatta, K. Shresta, and J. Y. Park, Ultrahigh-Output Triboelectric and Electromagnetic Hybrid Generator for Self-Powered Smart Electronics and Biomedical Applications, **Advanced Energy Materials**, (2022), pp. 2202238

7) S. Roy, M.H. Kabir, M. Salauddin, and M.A. Halim, An Electromagnetic Wind Energy Harvester Based on Rotational Magnet Pole-Pairs for Autonomous IoT Applications, **Energies**, Vol. 15, (2022), PP. 5725

8) M. Salauddin, SM S. Rana, M. Sharifuzzaman, S. Lee, M. A. Zahed, Y.D. Shin, S. Seonu, H.S. Song, T. Bhatta, and J. Y. Park, Laser-carbonized MXene/ZiF-67 nanocomposite as an intermediate layer for boosting the output performance of fabric-based triboelectric nanogenerator, **Nano Energy**, Vol. 100, (2022), PP. 107462

9) M. T. Rahman, SM S. Rana, M. Salauddin, M. A Zahed, S. Lee, E.S. Yoon, and J. Y. Park, Silicone-incorporated nanoporous cobalt oxide and MXene nanocomposite-coated stretchable fabric for wearable triboelectric nanogenerator and self-powered sensing applications, **Nano Energy**, Vol. 100 (2022), 107454

10) K. Shrestha, P. Maharjan, T. Bhatta, S. Sharma, M. T. Rahman, S. Lee, M. Salauddin, SM S. Rana, and J. Y. Park, A High-Performance Rotational Energy Harvester Integrated with Artificial Intelligence-Powered Triboelectric Sensors for Wireless Environmental Monitoring System, **Advanced Engineering Materials**, (2022), pp. 202200286

11) T. Bhatta, P. Maharjan, K. Shrestha, S. Lee, M. Salauddin, M. T. Rahman, SM S. Rana, S. Sharma, C. Park, S. H. Yoon, and J. Y. Park, A Hybrid Self-Powered Arbitrary Wave Motion Sensing System for Real-Time Wireless Marine Environment Monitoring Application, **Advanced Energy Materials**, Vol. 12 (2022), pp. 2102460

12) M. Salauddin, SM S. Rana, M. T. Rahman, Md. Sharifuzzaman, P. Maharjan, T. Bhatta, H. Cho, S.H. Lee, C. Park, K. Shrestha, S. Sharma, and J. Y. Park, Fabric-Assisted MXene/Silicone Nanocomposite-Based Triboelectric Nanogenerators for Self-Powered Sensors and Wearable Electronics, **Advanced Functional Materials**, Vol. 32 (2021), PP.2107143

13) MS Rasel, P. Maharjan, M. T. Rahman, M. Salauddin, SM. S. Rana, S. H. Lee, and J. Y. Park, Highly Responsive and Robust Micro-/Nano-Textured Self-Powered Triboelectric Humidity Sensor, **ACS Applied Electronic Materials**, Vol. 3 (2021), PP. 4376-4387

14) SM. S. Rana, M. A. Zahed, M. T. Rahman, M. Salauddin, S. H. Lee, C. Park, P. Maharjan, T. Bhatta, K. Shrestha, and J. Y. Park; Cobalt-nanoporous carbon functionalized nanocomposite-based triboelectric nanogenerator for contactless and sustainable self-powered sensor systems, **Advanced Functional Materials**, Vol. 31 (2021), pp. 2105110

15) SM S. Rana, M. T. Rahman, S. Sharma, M. Salauddin, S. H. Yoon, C. Park, P. Maharjan, T. Bhatta, and J. Y. Park, Cation functionalized nylon composite nanofibrous mat as a highly positive friction layer for robust, high output triboelectric nanogenerators and self-powered sensors, **Nano Energy**, Vol. 88 (2021), pp. 106300

16) M. T. Rahman, SM S. Rana, P. Maharjan, M. Salauddin, T. Bhatta, H. Cho, C. Park, and J. Y. Park, Ultra-robust and broadband rotary hybridized nanogenerator for self-sustained smart-farming applications, **Nano Energy**, Vol. 85 (2021), pp. 105974

17) P. Maharjan, T. Bhatta, C. Park, H. Cho, K. Shrestha, S Lee, M. Salauddin, M. T. Rahman, SM S. Rana, and J. Y. Park, High-Performance Keyboard Typing Motion Driven Hybrid Nanogenerator, **Nano Energy**, Vol. 88 (2021), pp. 106232

18) P. Maharjan, K. Shrestha, T. Bhatta, H. Cho, C. Park, M. Salauddin, M. T. Rahman, SM S. Rana, S Lee, and J. Y. Park, Keystroke Dynamics based Hybrid Nanogenerators for Biometric Authentication and Identification using Artificial Intelligence, **Advanced Science**, Vol. 8 (2021), 2100711

19) SM S. Rana, M. T. Rahman, M. Salauddin, S. Sharma, P. Maharjan, T. Bhata, H. Cho, C. Park, and J. Y. Park, Electrospun PVDF-TrFE/MXene Nanofiber Mat-Based Triboelectric Nanogenerator for Smart Home Appliances, **ACS Applied Materials & Interfaces**, Vol. 13 (2021), pp. 4955–4967

20) T. Bhata, P. Maharjan, M. Salauddin, M. T. Rahman, SM S. Rana, and J. Y. Park, High-Performance Triboelectric Nanogenerator Based on MXene Functionalized Polyvinylidene Fluoride Composite Nanofibers, **Nano Energy**, Vol. 81 (2021), pp. 105670

21) M. Salauddin, SM S. Rana, Md. Sharifuzzaman, M. T. Rahman, C. Park, H. Cho, P. Maharjan, T. Bhatta, and J. Y. Park, A Novel MXene/Ecoflex Nanocomposite-Coated Fabric as a Highly Negative and Stable Friction Layer for High-Output Triboelectric Nanogenerators, **Advanced Energy Materials**, (2020), pp. 2002832

22) P. Maharjan, T. Bhatta, H. Cho, X Hui, C. Park, S. Yoon, M. Salauddin, M. T. Rahman, SM S. Rana, and J. Y. Park, A Fully Functional Universal Self-Chargeable Power Module for Portable/Wearable Electronics and Self-Powered IoT Applications, **Advanced Energy Materials**, Vol. 10 (2020), pp. 2002782

23) M. T. Rahman, SM S. Rana, M. Salauddin, P. Maharjan, T. Bhatta, H. Kim, H. Cho, and J. Y. Park, A highly miniaturized freestanding kinetic-impact-based non-resonant hybridized electromagnetic-triboelectric nanogenerator for human induced vibrations harvesting, **Applied Energy**, Vol. 279 (2020), pp. 115799

24) P. Maharjan, T. Bhatta, M. Salauddin, M. S. Rasel, M. T. Rahman, SM S. Rana and J. Y. Park, A human skin-inspired self-powered flex sensor with thermally embossed microstructured triboelectric layers for sign language interpretation, **Nano Energy**, Vol. 76 (2020), pp. 105071

25) SM S. Rana, M. T. Rahman, M. Salauddin, P. Maharjan, T. Bhata, H. Cho and J. Y. Park, A human-machine interactive hybridized biomechanical nanogenerator as a self-sustainable power source for multifunctional smart electronics applications, **Nano Energy**, Vol. 76 (2020), pp. 105025

26) T. Bhata, P. Maharjan, M. Salauddin, M. T. Rahman, SM S. Rana, and J. Y. Park, A battery-less arbitrary motion sensing system using magnetic repulsion-based self-powered motion sensors and hybrid nanogenerator, **Advanced Functional Materials**, (2020), pp. 2003276

27) M. T. Rahman, SM S. Rana, M. Salauddin, P. Maharjan, T. Bhata, and J. Y. Park, Biomechanical energy-driven hybridized generator as a universal portable power source for smart/wearable electronics, **Advanced Energy Materials**, Vol. 2020 (2020), pp. 1903663

28) P. Maharjan, T. Bhatta, M. S. Rasel, M. Salauddin, R. M. Toyabur, and J. Y. Park, High-performance cycloid inspired wearable electromagnetic energy harvester for scavenging human motion energy, **Applied Energy**, Vol. 256 (2019), pp. 113987

29) M. Salauddin, R. M. Toyabur, P. Maharjan, M. S. Rasel, H. Cho, and J. Y. Park, Design and experimental analysis of a low-frequency resonant hybridized nanogenerator with a wide bandwidth and high output power density, **Nano Energy**, Vol. 66 (2019), pp. 104122

30) J. W. kim1, M. Salauddin1, H. Cho, M. S. Rasel, and J. Y. Park, Electromagnetic energy harvester based on a finger trigger rotational gear module and an array of disc Halbach magnets, **Applied Energy**, Vol. 250 (2019), pp. 776-785 1These authors contributed equally to this work.

31) R. M. Toyabur, M Salauddin, P. Maharjan, M. S. Rasel, H. Cho, and J. Y. Park, Natural wind-driven ultra-compact and highly efficient hybridized nanogenerator for self-sustained wireless environmental monitoring system, **Nano Energy**, Vol. 57 (2019), pp. 256-268

32) J. Y. Park, M. Salauddin, and M. S. Rasel, Nanogenerator for scavenging low frequency vibrations, **Journal of Micromechanics and Microengineering**, Vol. 29 (2019), pp. 053001

33) P. Maharjan, M. Salauddin, H. Cho, and J. Y Park, An indoor power line based magnetic field energy harvester for self-powered wireless sensors in smart home applications, **Applied Energy**, Vol. 232 (2018), pp. 398-408

34) M. Salauddin, M. T. Rahman, P. Maharjan, M. S. Rasel, J. W. Kim, H. Cho, and J. Y. Park, Miniaturized springless hybrid nanogenerator for powering portable and wearable electronic devices from human-body-induced vibration, **Nano Energy**, Vol. 51 (2018), pp. 61-72

35) P. Maharjan, H. Cho, M. S. Rasel, M. Salauddin, and J. Y. Park, A fully enclosed, 3d printed, hybridized nanogenerator with flexible flux concentrator for harvesting diverse human biomechanical energy, **Nano Energy**, Vol. 53 (2018), pp. 213-224

36) M. T. Rahman, M. Salauddin, H. Cho, and J. Y. Park, A multimodal hybrid energy harvester based on piezoelectric-electromagnetic mechanisms for low-frequency ambient vibrations, **Energy Conversion and Management**, vol. 168 (2018), pp. 454-456

37) M. S. Rasel, P. Maharjan, M. Salauddin, R. M. Toyabur, H. Cho, J. W. Kim, and J. Y. Park, An impedance tunable and highly efficient triboelectric nanogenerator for large-scale, ultra-sensitive pressure sensing applications, **Nano Energy**, Vol. 49 (2018), pp. 603-613

38) M. Salauddin, R. M. Toyabur, P. Maharjan, and J. Y Park, High performance human-induced vibration driven hybrid energy harvester for powering portable electronics, **Nano Energy**, Vol. 45 (2018), PP. 236-246

39) M. Salauddin, H. Cho, and J. Y. Park, A hybrid electromagnetic-triboelectric energy harvester using a dual Halbach magnet array powered by human-body-induced motion, Advanced Materials Technologies, Vol. 3 (2018), PP. 1700240

40) M. Salauddin, M. S. Rasel, J. W. Kim, and J. Y. Park, Design and experiment of hybridized electromagnetic-triboelectric energy harvester using Halbach magnet array from handshaking vibration, **Energy Conversion and Management**, Vol. 153 (2017), PP. 1-11

41) M. Salauddin and J. Y. Park, Design and experiment of human hand motion driven electromagnetic energy harvester using dual Halbach magnet array, **Smart Materials and Structures**, vol. 26 (2017), pp. 035011

42) R. M. Toyabur, M. Salauddin, and J. Y. Park, Design and experiment of piezoelectric multimodal energy harvester for low frequency vibration, **Ceramics International**, Vol. 43(2017), PP. S675-S681

43) M. A. Halim, H. Cho, M. Salauddin, and J. Y. Park, A miniaturized electromagnetic vibration energy harvester using flux-guided magnet stacks for human-body-induced motion, **Sensors and Actuators A: Physical**, Vol. 249 (2016), pp.23 -31

44) M. Salauddin, M. A. Halim, and J. Y. Park, A magnetic-spring-based, low-frequency vibration energy harvester comprising a dual Halbach array, **Smart Materials and Structures**, Vol. 25 (2016), pp. 095017

CONFERENCE PRESENTATIONS (14)

1) M. Salauddin, P. Gupta, Y. Fu, M Hernadez, Y. C. Wang, Food-Waste Derived Triboelectric Sensors for Biomechanical Monitoring of Frailty Status and Fall Risk in Older Adults, 2025 Design of Medical Devices Conference, April 28–30, 2025. [International]

2) M. Salauddin, O. Faruk, SM Rana, Y.Y. Lee, and J. Y. Park, V2CTX Functionalized Self-Powered Triboelectric Sensors for Sign Language Interpretation, 18th IEEE International Conference on Nano/Micro Engineered and Molecular Systems (NEMS), December 14-17, Jeju, South Korea, 2023 [International, Oral]

3) SM Rana, M. Salauddin, M. Toyabur, S. Yoon, H. Cho and J. Y. Park, A Multilayer Stretchable Triboelectric Nanogenerator Based on Metal-Organic Framework for Wearable Self-Powered Biomotion and Tactile Sensors, 21st International Conference on Micro and Nanotechnology for Power Generation and Energy Conversion Applications (PowerMEMS), December 12-15, Utah, USA, 2022, pp. 22-25 [International, Oral]

4) SM Rana, M. Salauddin, M. Toyabur, S. Yoon, H. Cho and J. Y. Park, A Poly-DADMAC Functionalized Nanofibours Mat-Based Self-Powered Human Motion Sensor for IoT Applications, The 21st International Conference on Solid-State Sensors, Actuators and Microsystems (Transducers-2021), June 20-25, 2021, pp. 569-572 [International, Oral]

5) SM Rana, M. Toyabur, M. Salauddin, H. Cho, and J. Y. Park, An Electrospun PVDF-TRFE/Mxene Nanofibours Mat-Based Self-Powered Motion Sensor, The 34th IEEE International Conference on Micro Electro Mechanical Systems (MEMS-2021), January 25-29, 2021, pp. 30-33 [International, Oral]

6) R. M. Toyabur, M. Salauddin, and J. Y. Park, A natural wind-driven 3d-printed miniaturized and fully enclosed hybrid nanogenerator using flexible blade structure for subway tunnel applications, Transducers-2019, June 23-27, Berlin, Germany, pp. 1443-1446 [International, Poster]

7) M. Salauddin, R. M. Toyabur, and J. Y. Park, A human motion driven electromagnetic and triboelectric hybridized nanogenerator for powering portable electronics, The 4th International Conference on Nanogenerators and Piezotronics (NGPT-2018), May 8-11, 2018, Sungkyunkwan University, Korea [International, Poster]

8) M. Salauddin, R. M. Toyabur, and J. Y. Park, A free motion driven electromagnetic and triboelectric hybridized nanogenerator for scavenging low frequency vibrations, The 31st IEEE International Conference on Micro Electro Mechanical Systems (MEMS-2018), January 21-25, 2018, Belfast, Northern Ireland, UK, pp. 233-236 [International, Oral]

9) M. Salauddin and J. Y. Park, A low frequency vibration driven, miniaturized and hybridized electromagnetic and triboelectric energy harvester using dual Halbach array, The 19th International Conference on Solid-State Sensors, Actuators and Microsystems (Transducers-2017), June 18-22, 2017, Kaohsiung, Taiwan, pp. 1832-1835 [International, Poster]

10) M. Salauddin and J. Y. Park, A handy motion driven hybrid energy harvester: dual Halbach array based electromagnetic and triboelectric generators, The 16th International Conference on Micro- and Nano-Technology for Power Generation and Energy Conversion Applications (PowerMEMS-2016), December 6-9, 2016, Paris, France, pp. 012004 [International, Oral]

11) M. Salauddin, M. A. Halim, M. S. Rasel, and J. Y. Park, A halbach array based hybridized electromagnetic-triboelectric nanogenerator for powering portable electronics, The 14th International nanotech symposium & nano-convergence Expo, NanoKorea-2016, July 13-15, 2016, KINTEX, South Korea, pp. 55 [International, Oral]

12) M. Salauddin, M. A. Halim, and J. Y. Park, A low frequency vibration energy harvester using dual halbach array suspended in magnetic springs, The 15th International Conference on Micro- and Nano-Technology for Power Generation and Energy Conversion Applications (PowerMEMS-2015), December 1-4, 2015, MIT, Boston, USA, pp. 012011 [International, Oral]

13) M. Salauddin, M. A. Halim, and J. Y. Park, A Low Frequency Vibration Electromagnetic Energy Harvester Using Dual Halbach Magnets Array, KMEMS-2016, April 7-9, 2016, Jeju, South Korea, pp. 469-470 [Domestic, Poster]

14) M. Salauddin, C. Cha, and J. Y. Park, Efficiency improvement of wireless power transfer system by using mutual coupling resonance technique, KMEMS-2015, April 2-4, 2015, Jeju, South Korea, pp. 373-374 [Domestic, Poster]

PATENTS

Electromagnetic harvester using magnet springs, Registration No. 10-2017-0000857, 2017.01.03

Hybrid electromagnetic and sliding energy harvester using multiple magnets and PDMS arranged with double halbach array, Registration No. 10-2017-0097050, 2017.07.31

Non-resonant high power hybrid energy harvester, Registration No. 10-2018-0078849, 2018.07.06

AWARDS & SCHOLARSHIPS

Brain Pool Fellowship (BPF), National Research Foundation of Korea, 2020–2023

Best Poster Award, International Conference on Nanogenerators and Piezotronics, 2018

Student Travel Grant, IEEE MEMS 2018, Transducer Research Foundation, UK, 2018

Graduate Research Scholarship, Kwangwoon University, South Korea, 2014–2016

EDITORIAL & PROFESSIONAL SERVICE

Energies Journal: Special issue on “Low-Frequency Vibration-Based Electromagnetic Energy Harvesters” Energies (ISSN 1996-1073), MDPI, SWITZERLAND.

Applied Sciences Journal: Special issue on “Advances of Triboelectric Nanogenerators for Energy Harvesting and Sensors Applications” Applied Sciences (ISSN 2076-3417), MDPI, SWITZERLAND.

Electronics Journal: Special issue on “Latest Advances in Energy Harvesting Technologies and Applications” Electronics (ISSN 2079-9292), MDPI, SWITZERLAND.

Peer reviewer for ACS OMEGA, Smart Materials and Structures, Journal of Physics D: Applied Physics, Measurement Science and Technology, Physica Scripta, Engineering Research Express, Applied Physics Letters, IEEE Access, IEEE Sensors, Nanomaterials, Energies, Biosensors, Micromachines, Electronics, Actuators, Coatings, Materials, Sustainability, and Sensors.